

The one-dimensional endpoint Hardy–Lieb–Thirring inequality on arbitrary domains

Autonomous proof attempt for arXiv:1102.4394

11 August 2026

Abstract

Frank and Loss proved a Hardy–Lieb–Thirring inequality on arbitrary domains in dimension one for moments $\gamma > 1/2$ and wrote that the endpoint $\gamma = 1/2$ should also hold. We prove that endpoint. A finite interval is split at its midpoint. The two Dirichlet halves are controlled by the known critical half-line endpoint inequality. Removing the midpoint condition is a codimension-one form extension, so interlacing leaves only the lowest eigenvalue. The source paper’s pointwise Hardy–Sobolev inequality controls that eigenvalue. Summing over interval components gives a universal constant, explicitly $C_{1/2,0} + 2 \leq 3.185$ in terms of the published half-line constant.

1 Statement

Let $\Omega \subsetneq \mathbb{R}$ be open and put

$$d_\Omega(x) = \text{dist}(x, \Omega^c).$$

In dimension one the averaged distance D_Ω in the source paper equals d_Ω . Let $H_{\Omega,V}$ be the Friedrichs operator associated with the quadratic form

$$\mathfrak{q}_{\Omega,V}[u] = \int_\Omega \left(|u'|^2 - \frac{|u|^2}{4d_\Omega^2} + V|u|^2 \right) dx. \quad (1)$$

The source proves $\text{Tr}(H_{\Omega,V})_-^\gamma \leq L_{1,\gamma} \int_\Omega V_-^{\gamma+1/2}$ for $\gamma > 1/2$ and asks immediately after Theorem 1.5 whether it remains true at $\gamma = 1/2$.

Theorem 1. *There is a universal constant $L_{1,1/2}$ such that, for every open $\Omega \subsetneq \mathbb{R}$ and every real $V \in L^1(\Omega)$,*

$$\text{Tr}(H_{\Omega,V})_-^{1/2} \leq L_{1,1/2} \int_\Omega V_-(x) dx. \quad (2)$$

One may take

$$L_{1,1/2} = C_{1/2,0} + 2 \leq 3.185,$$

where $C_{1/2,0} \leq 1.185$ is the upper bound in the critical half-line inequality of Ekholm and Frank.

2 Two inputs

The first input is Ekholm–Frank, Theorem 2.1 in arXiv:math/0611247. In the special case $\alpha = 0$, $\gamma = 1/2$ it says that, for $W \geq 0$ on $\mathbb{R}_+ = (0, \infty)$,

$$\text{Tr} \left(-\frac{d^2}{dr^2} - \frac{1}{4r^2} - W \right)_-^{1/2} \leq C_{1/2,0} \int_0^\infty W(r) dr, \quad (3)$$

with the Dirichlet condition at the singular endpoint.

The second input is Corollary 2.2 of the source paper at $q = 2$. If I is an interval and

$$\mathfrak{h}_I[u] = \int_I \left(|u'|^2 - \frac{|u|^2}{4d_I^2} \right) dx, \quad d_I(x) = \text{dist}(x, I^c), \quad (4)$$

then

$$\|u\|_\infty^4 \leq 16 \mathfrak{h}_I[u] \|u\|_2^2. \quad (5)$$

The source states the bound first on $C_c^\infty(I)$; it extends to the closed form domain by density. In particular, evaluation at an interior point is continuous in the form norm.

3 A finite interval

Fix $I = (a, b)$, let $m = (a + b)/2$, and write $W = V_- \geq 0$ and $A_I = \int_I W$. Since positive parts of V only raise the operator, it suffices to study

$$H_I = -\frac{d^2}{dx^2} - \frac{1}{4d_I(x)^2} - W.$$

Let H_I^D be the form restriction obtained by imposing $u(m) = 0$. It is the orthogonal direct sum of the operators on (a, m) and (m, b) with a Dirichlet condition at m . On the left half $d_I(x) = x - a$, and on the right half $d_I(x) = b - x$. After translation or reflection, each half is therefore the critical half-line form on $(0, R)$, $R = (b - a)/2$, with an extra Dirichlet condition at R .

Zero extension at R embeds this finite-interval Dirichlet form domain into the half-line form domain. Extend the corresponding half-potential by zero beyond R . The min-max principle and (3), applied to both halves, give

$$\text{Tr}(H_I^D)_-^{1/2} \leq C_{1/2,0} A_I. \quad (6)$$

The condition $u(m) = 0$ has codimension one. If $\mu_1 \leq \mu_2 \leq \dots$ and $\nu_1 \leq \nu_2 \leq \dots$ are respectively the negative eigenvalues of H_I and H_I^D , with the lists continued by the bottom of the nonnegative spectrum when needed, form interlacing gives

$$\mu_j \leq \nu_j \leq \mu_{j+1}. \quad (7)$$

Consequently

$$\text{Tr}(H_I)_-^{1/2} \leq (-\mu_1)_+^{1/2} + \text{Tr}(H_I^D)_-^{1/2}. \quad (8)$$

It remains to estimate the single possible extra eigenvalue. For $\|u\|_2 = 1$, (5) implies

$$\begin{aligned} \mathfrak{h}_I[u] - \int_I W |u|^2 &\geq \mathfrak{h}_I[u] - A_I \|u\|_\infty^2 \\ &\geq \mathfrak{h}_I[u] - 4A_I \mathfrak{h}_I[u]^{1/2} \geq -4A_I^2. \end{aligned}$$

Thus $\mu_1 \geq -4A_I^2$, and (6)–(8) yield

$$\text{Tr}(H_I)_-^{1/2} \leq (C_{1/2,0} + 2) A_I. \quad (9)$$

4 Arbitrary open sets

Every open proper subset of $\mathbb{R}$ is a countable disjoint union of finite intervals and rays. On a ray the distance to the complement is exactly the distance to its finite endpoint, so (3) applies directly and gives the stronger constant $C_{1/2,0}$. On every finite component use (9). The operator on Ω is the orthogonal direct sum of the component operators. Summing their eigenvalue moments gives

$$\begin{aligned}\mathrm{Tr}(H_{\Omega,-V_-})_-^{1/2} &\leq (C_{1/2,0} + 2) \sum_I \int_I V_- \\ &= (C_{1/2,0} + 2) \int_{\Omega} V_-.\end{aligned}$$

This proves (2) for bounded compactly supported V_- . Monotone truncation and standard form convergence give the result for every $V_- \in L^1(\Omega)$, and monotonicity restores the positive part of V . $\square$

Scope and source note

This settles the explicit $N = 1$, $\gamma = 1/2$ sentence on source PDF page 5. It does not settle the paper's separate question about a p -Hardy–Sobolev inequality for $1 < p < 2$. The proof uses only the source's Corollary 2.2 and Tomas Ekholm and Rupert L. Frank, *Lieb–Thirring inequalities on the half-line with critical exponent*, arXiv:math/0611247, Theorem 2.1.